Andres Flores Nieto

Mechatronics Engineer

andres.flores.original@gmail.com LinkedIn Profile México, Puebla Portfolio on Github

Profile

Dynamic and innovative professional with experience for their leadership abilities and participation in national competitions, demonstrating a commitment to continuous improvement and a passion for developing impactful technological solutions.

Experience History

- Participate in the VEX Robotics Competition

Working in a team, you addressed complex robotic challenges using creative and unconventional design solutions. This experience was key to developing skills in prototyping, technical problem-solving, and critical thinking, which were crucial for creating a unique and functional robot.

- Participate in competitions First (For Inspiration and Recognition of Science and Technology)

Collaborated with a multidisciplinary university team to design, build, and program a robot for a competitive challenge. By applying advanced engineering principles, you optimized the robot's performance to achieve superior results. This experience highlights your agile problem-solving skills and strong commitment to meeting project goals.

- Participate in the Innovation Meetup Competition

Led a university team to solve a real-world business problem. Using a structured approach, you guided the team from ideation to the development of a creative, implementable solution. This demonstrates your strong capacity for identifying core issues and developing innovative strategies that align with business objectives.

Projects

- 6 Degree of Freedom Robotic Arm

A 6 DOF robotic arm was developed starting from a conceptual design evaluated among several alternatives. The selected model was designed in CAD, manufactured by 3D printing, and assembled with the necessary electronic components. The system was programmed in Arduino to control each joint of the arm individually.

- Mobile robotic platform controlled by Bluetooth and Wi-Fi

A mobile carriage was developed using an ESP-32 board as the main control system. Electric motors, power electronics, and a wireless communication system were integrated. The robot was programmed to receive commands from an app via Bluetooth and also from a web interface via Wi-Fi. Connectivity, motion control, and system response tests were performed.

- Conveyor belt programmed with PLC

A small-scale conveyor belt was built using basic materials and electromechanical integration. The system was programmed using ladder logic to control start, stop, actuator activation, emergency stop, and operational light signaling. The system demonstrated functional operation by transporting and filling a glass using a motorized mechanism.

- Zumo-type robot

A Zumo-type robot was developed from a CAD design that was then manufactured by 3D printing. The system integrates control electronics, actuators, and sensors for environment perception. The robot's behavior was programmed and an operating strategy was defined for its operation.

- Line follower carriage

A mobile carriage was developed whose structural design was modeled in CAD and manufactured by 3D printing. The system integrates line detection sensors, control electronics, and actuators for movement. The system was programmed to adjust the vehicle's trajectory and maintain continuous line following at constant speed.

Education

Currently studying Mechatronics Engineering at TecMilenio University

Certificates

Solid Edge Certified Associate Level I

Certificate in Public Leadership

Certificate in Additive Manufacturing (in course)

Knowledge

Foundry & Casting Fundamentals

Additive Manufacturing

Advanced Excel

Office Suite

Python programming

C++ programming

Skilled Tinkercad

Skilled Solid Edge

Skilled Siemens NX

Autodesk Fusión Fundamentals

Electronic circuit simulation using NI Multisim

PLC programming

Programming in LabView

Fundamental Welding Skills

Software CAM Fundamentals

CNC Machining

Turning Processes

Electronic Circuit Prototyping & Assembly

Tecnomatix Plant Simulation Fundamentals

Basic GitHub repository management

Soft Skills

Teamwork

Leadership

Responsibility

Problem-solving

Creativity

Adaptability

Initiative in different challenges

Time management

Languages

Spanish - Native

English - C1